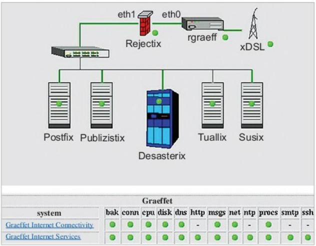


Figure 4: Part of an analysis done by the Big Sister tool (Image credits: *http://www.computerworlduk.com/*)

**Security Onion:** We should all be aware that network security monitoring is made up of many layers, just like an onion. Hence, no single tool will give us visibility into each and every attack or show us every reconnaissance or foot-print session on our company network. Security Onion actually bundles scores of different proven tools into one handy Ubuntu distro that allows us to see who is inside our network and helps keep the bad ones out. Whether we are taking a proactive approach to network security monitoring or even if we are following up on an attack, Security Onion can assist us.

Consisting of server, sensor and display layers, Security Onion combines full network packet capture with network-based and host-based intrusion detection. The network security toolchain also includes Netsniff-NG for packet capture; Suricata and Snort for rules-based network intrusion detection; OSSEC for host intrusion detection; Bro for analysis-based network monitoring; and Sguil, Snorby, Squert and ELSA (Enterprise Log Search and Archive) for display, analysis and log management. It's actually a collection of tools, all wrapped into a wizard-driven installer and backed by thorough documentation that can help us get complete network monitoring as fast as possible.

*Functionality and usage*

It helps in:

- Combining full network packet capture with the network-based and host-based intrusion detection
- Serving all different logs for inspection and analysis
- Analysis-based network monitoring
- Log management

**Big Sister:** This tool was created by Thomas Abey, since he was really impressed by the network monitoring functions performed by another such tool called Big Brother. Thomas wanted to improve the performance of the tool and reduce the number of alarms when some system goes down, while making other enhancements. Big Sister uses Node Director, Deoxygen Filter and Big Sister Web application frameworks in order to work as part of different UNIX derivatives and Microsoft Windows versions.

*Functionality and usage*

It helps in:

- Notifying admins when the system is in a critical state
- Generating history of status changes and logs
- Displaying a variety of system performance data

## Benefits of network monitoring

Let's look at the benefits that organisations get out of network monitoring and management.

1. It helps in optimising the availability and performance of any network that is being monitored.
2. Network monitoring also helps in lowering the expenses of any organisation implementing it by improving asset utilisation.
3. It minimises the risks associated with the whole system by providing a secure network which meets compliance guidelines
4. Monitoring of a network also ensures effective change management so that users can establish the solid baselines for its performance.
5. With optimal asset utilisation achieved by network monitoring, the terms of service level agreements are met and it is possible to document the performance using reports. END

### References

Network Monitoring and Management by Esad Saitović and Ivan Ivanović.

[1] *http://www.wikipedia.org/*
[2] *http://www.computerworlduk.com/*
[3] *http://www.infoworld.com/*

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